

Introduction to Computer Vision



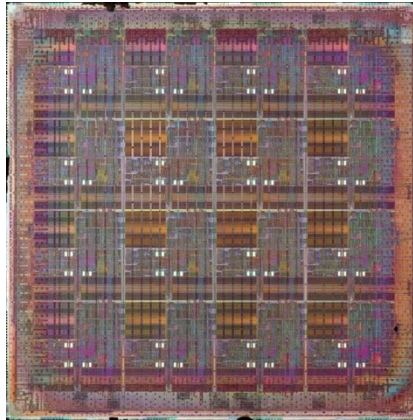
Lec.12: Computational Photography
2023.12.25

Computational Photography

- Data recorded by sensor is not the final image
- Computational Photography: arbitrary computation to the final image



Generalized imaging



Lots of computation



Final image

Today

High Dynamic Range Imaging (HDR)

Deblurring

Colorization

Super Resolution

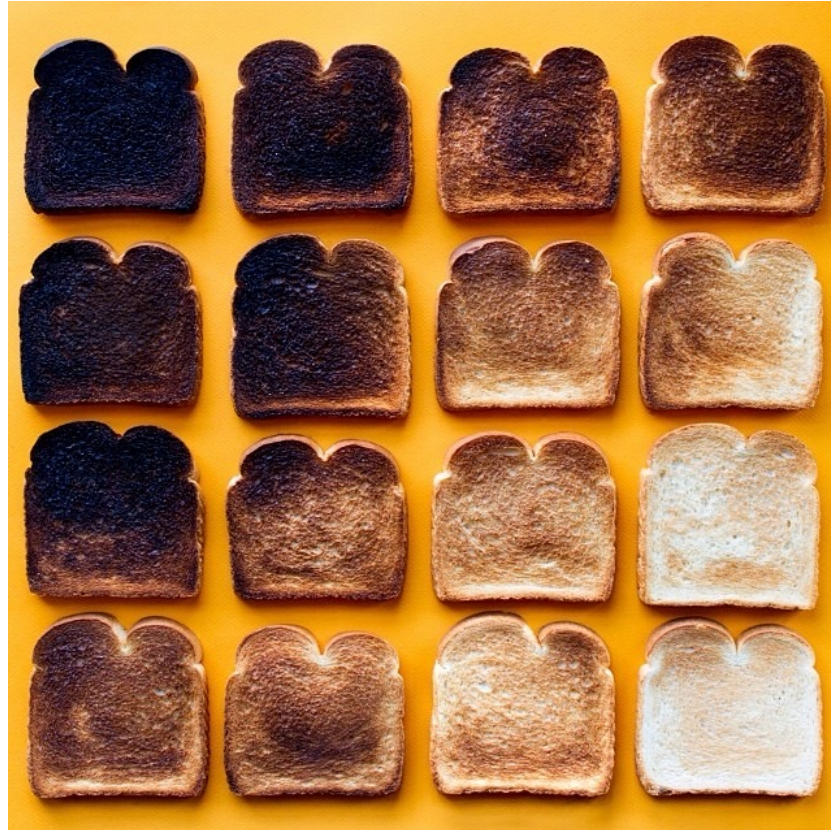
High dynamic range imaging (HDR)



High dynamic range imaging (HDR)



Exposure



Slides adapt from
graphics.cs.cmu.edu/courses/15-463, 20 Fall
stellar.mit.edu/S/course/6/sp15/6.815

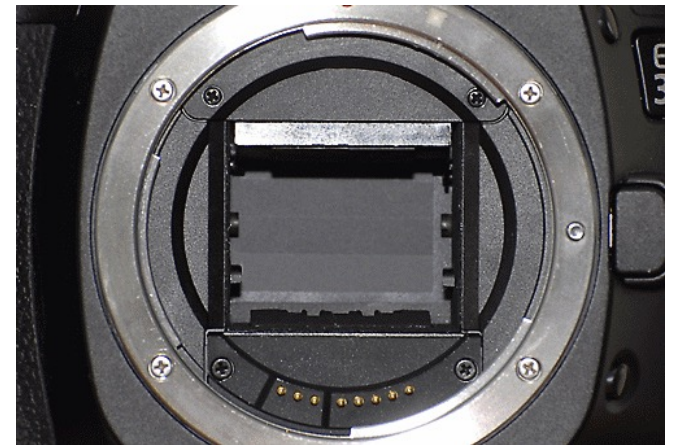
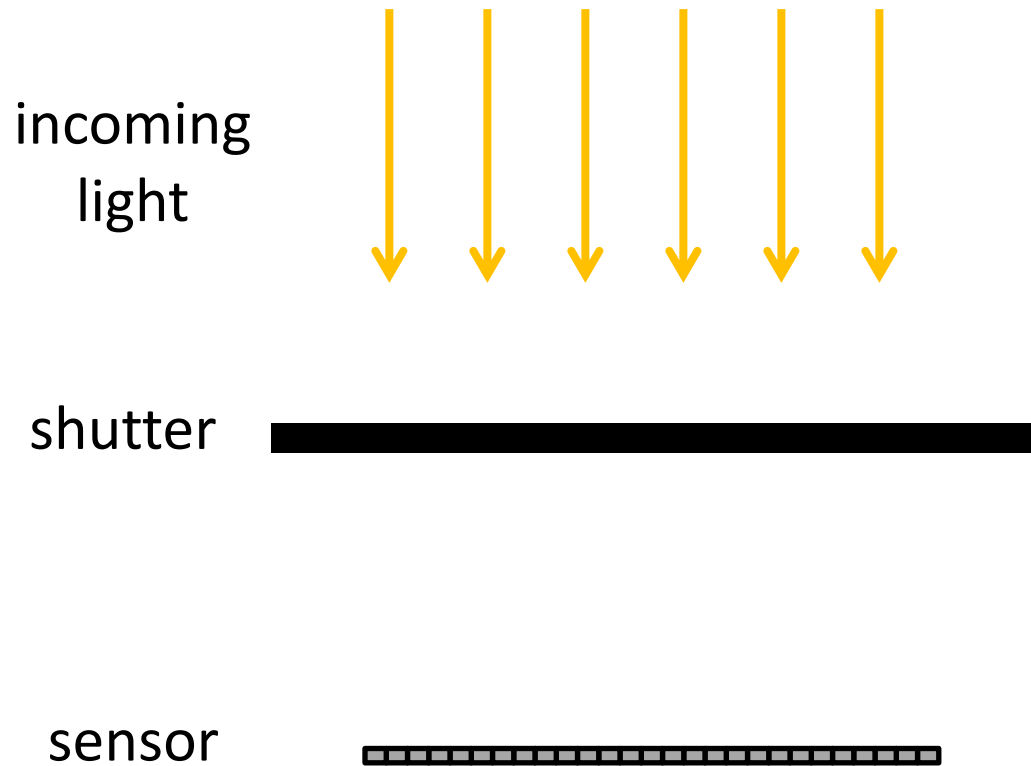
Exposure

Roughly speaking, the “brightness” of a captured image given a fixed scene.

$$\text{Exposure} = \text{Gain} \times \text{Irradiance} \times \text{Time}$$

- Gain is controlled by the ISO.
- Irradiance is controlled by the aperture.
- Time is controlled by the shutter speed.

Exposure: shutter speed



closed shutter

Exposure: aperture



Exposure: ISO

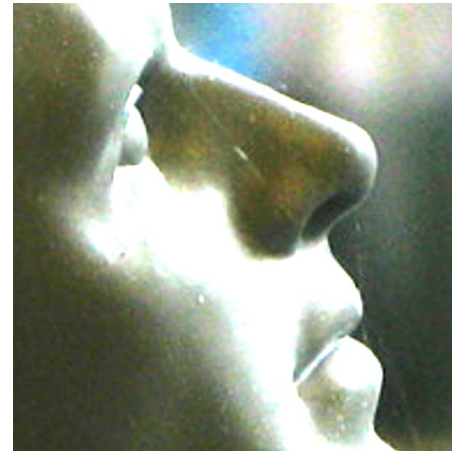
keep shutter speed and aperture **unchanged**



ISO 80



ISO 200



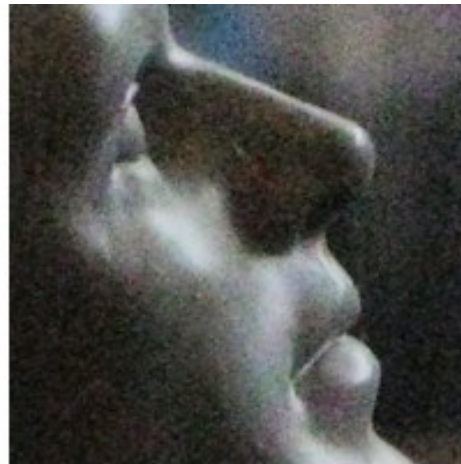
ISO 500

Exposure: ISO

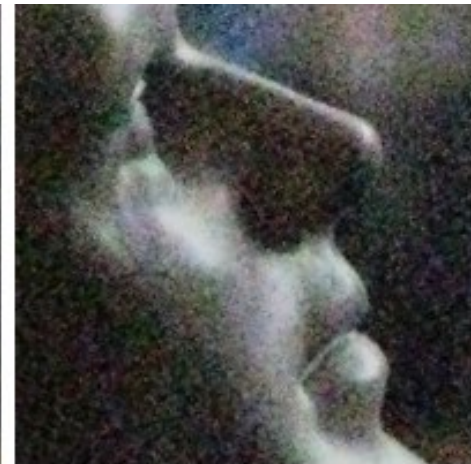
keep the final exposure **unchanged**



ISO 80



ISO 800

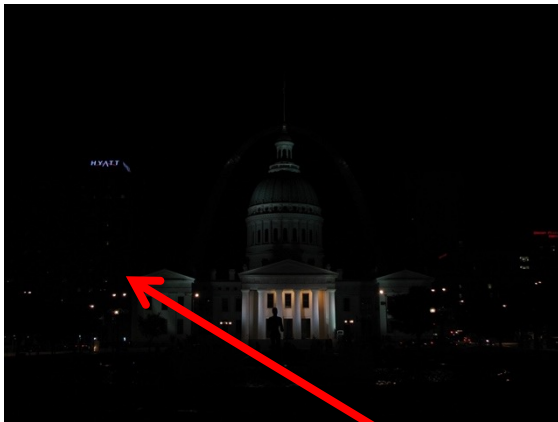


ISO 3200

Side-effects of ISO: Image becomes very grainy because noise is amplified.

Exposure = Gain x Irradiance x Time

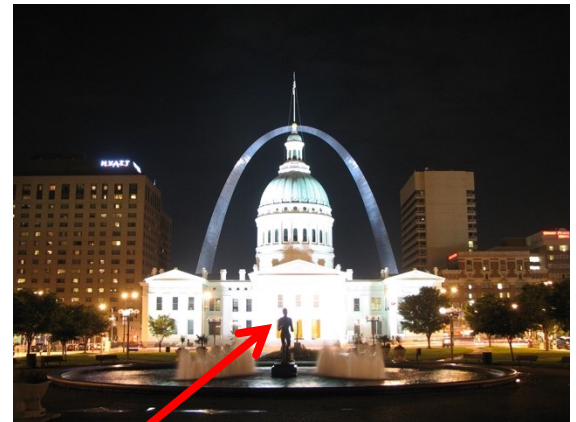
When taking a photo, **the averaged exposure** should be **at the middle of the sensor's measurement range**. So that the photo has both bright and dark parts with details.



underexposure



good exposure
with this sensor



overexposure

No details

Dynamic range

The ratio between the largest and smallest values of a certain quantity (e.g., brightness)

10:1	photographic print (higher for glossy paper)
256:1	8-bit RGB image
1000:1	LCD display
4096:1	digital SLR (at 12 bits)
100000:1	real world

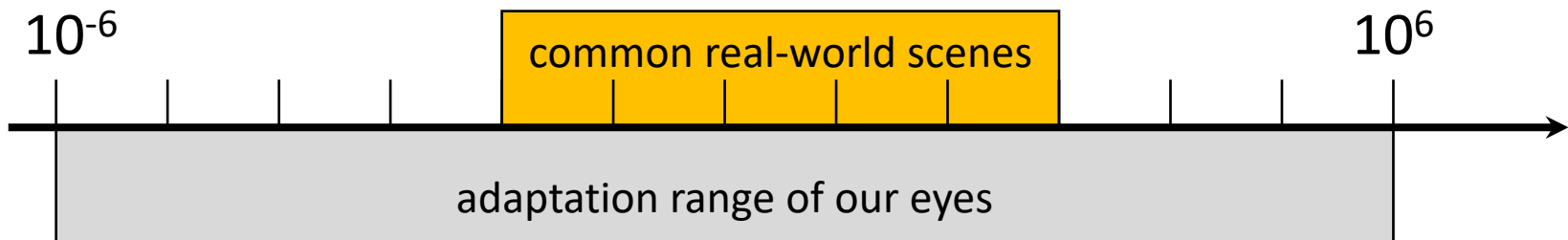
Dynamic range

The ratio between the largest and smallest values of a certain quantity (e.g., brightness)

Real world have a high dynamic range

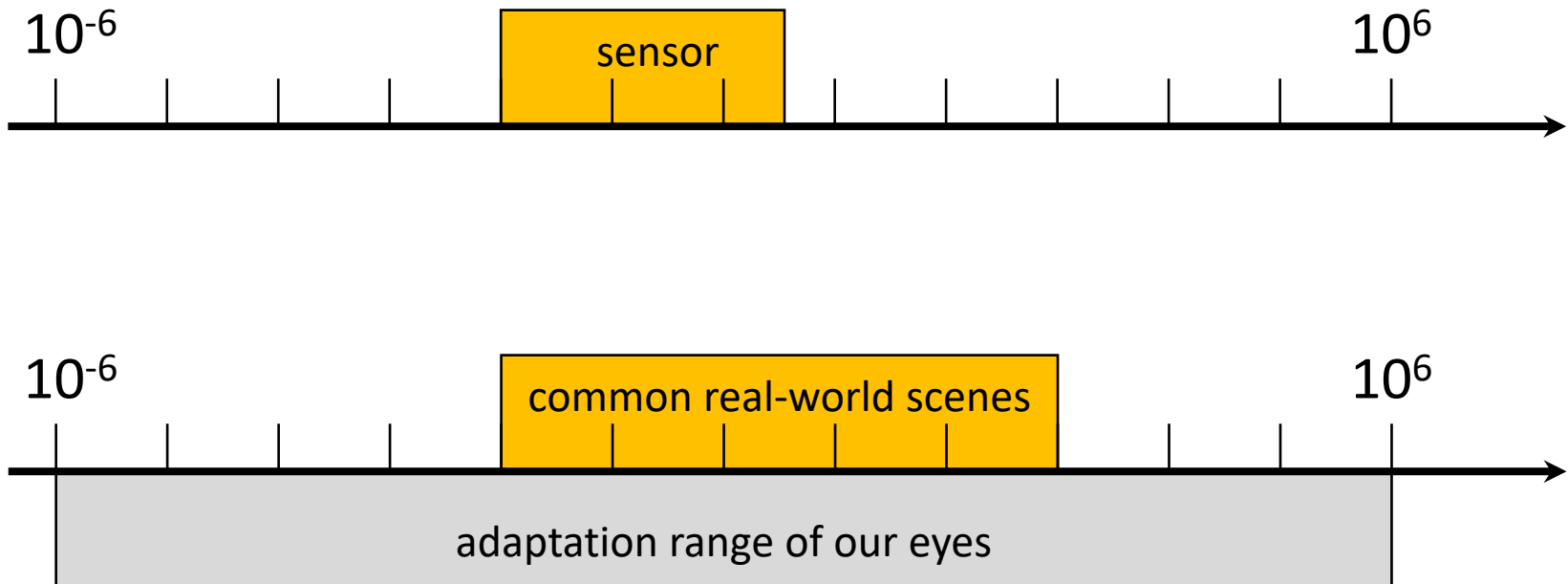


Figure 10.11 *Relative brightness of different scenes, ranging from 1 inside a dark room lit by a monitor to 2,000,000 looking at the Sun. Photos courtesy of Paul Debevec.*



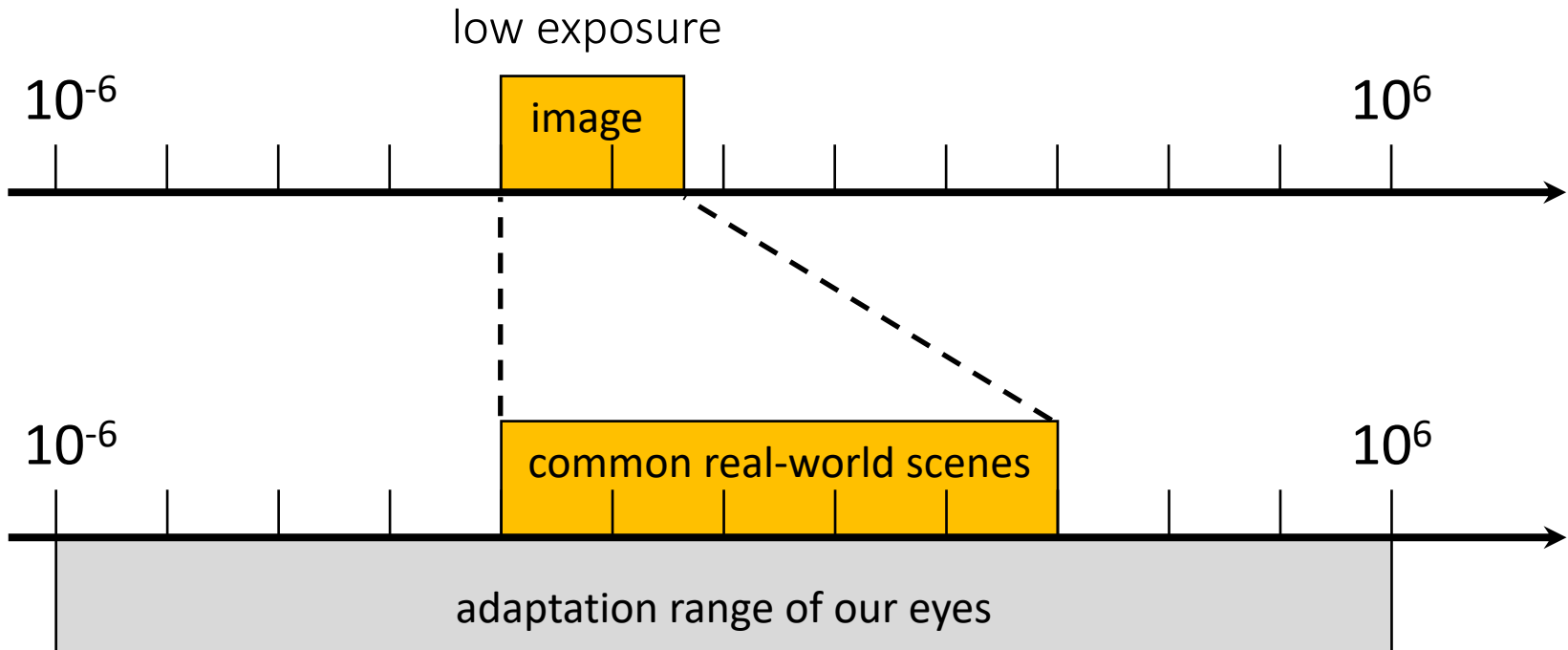
Dynamic range

Challenge: sensor's measurement range (dynamic range) does not match the world!



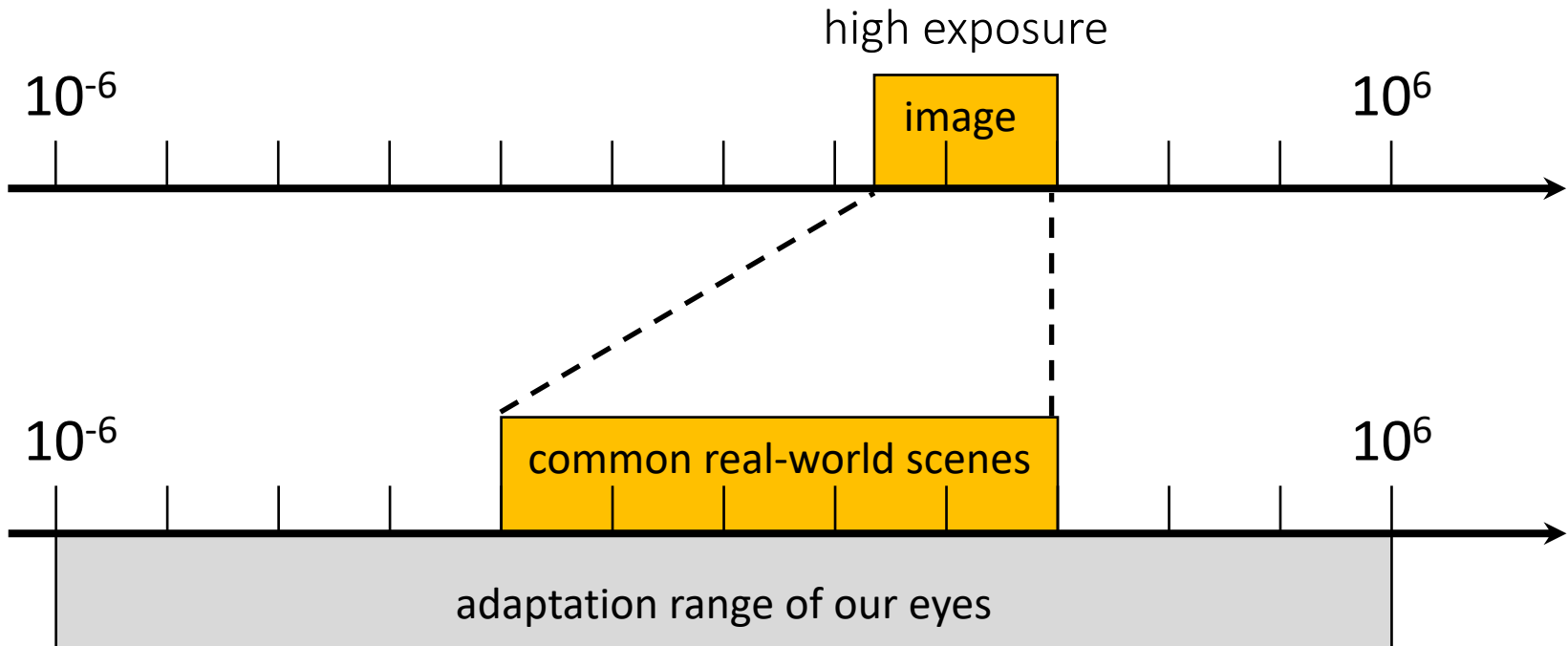
Dynamic range

Images have an even lower dynamic range
because of quantization (8-bit = 256)



Dynamic range

Images have an even lower dynamic range
because of quantization (8-bit = 256)



Without HDR

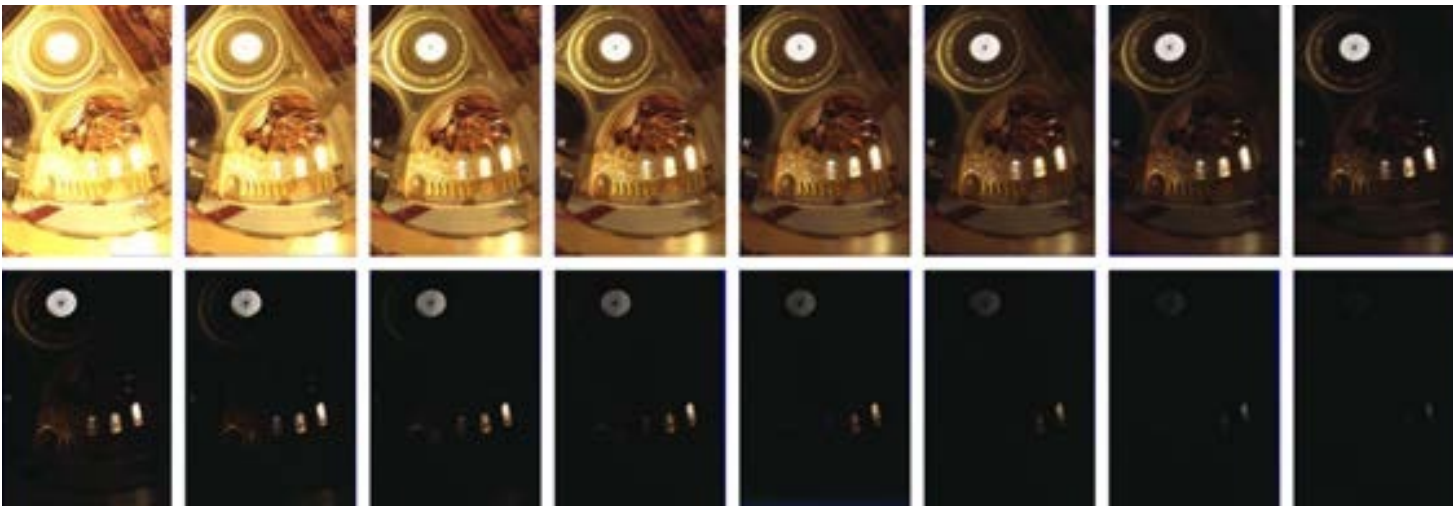


With HDR



Key idea

1. Exposure bracketing: Capture multiple LDR images at different exposures



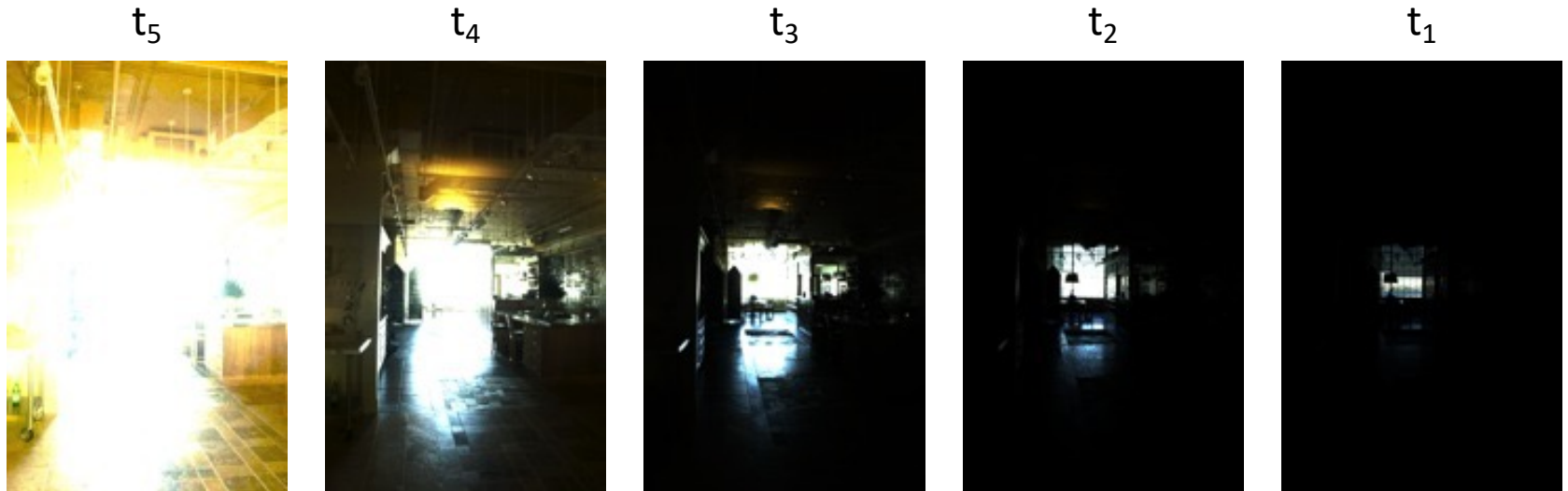
2. Merging: Combine them into a single HDR image



Image formation model

Suppose scene radiance for image pixel (x, y) is $L(x, y)$

Exposure time:



What is an expression for the image $I(x, y)$ as a function of $L(x, y)$?

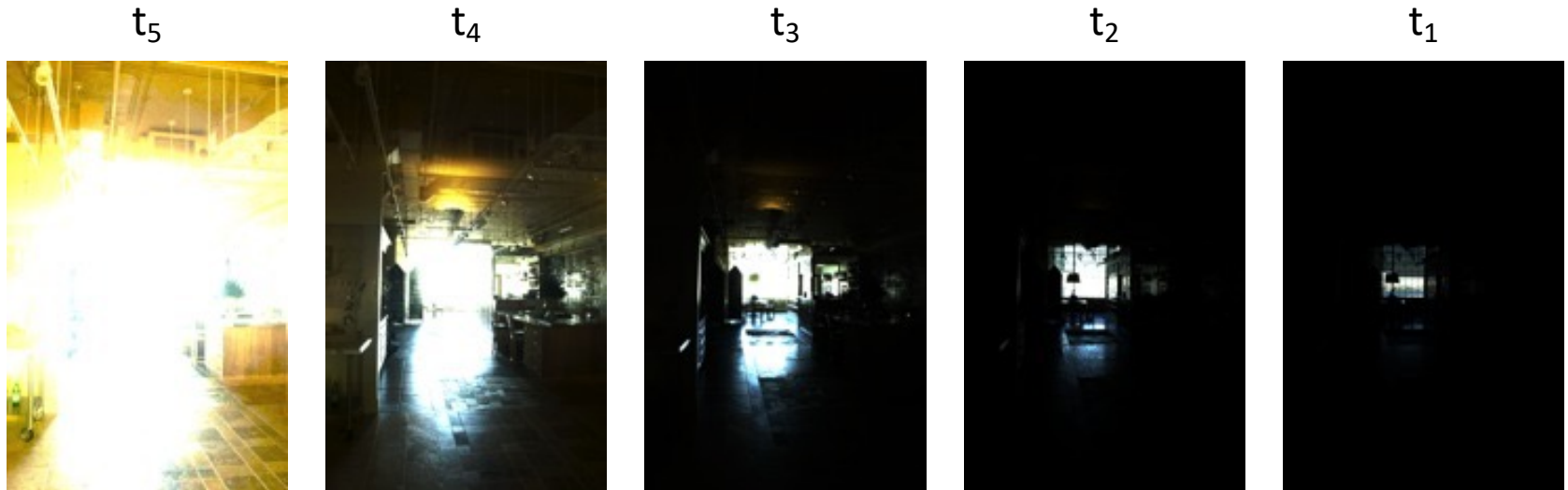
$$I(x, y) = \text{clip}[t_i \cdot L(x, y) + \text{noise}]$$

Merging images

For each pixel:

1. Find “valid” pixels in each image
2. Weight valid pixel values appropriately
3. Form a new pixel value as the weighted average of valid pixel values

How would you implement steps 1-2?

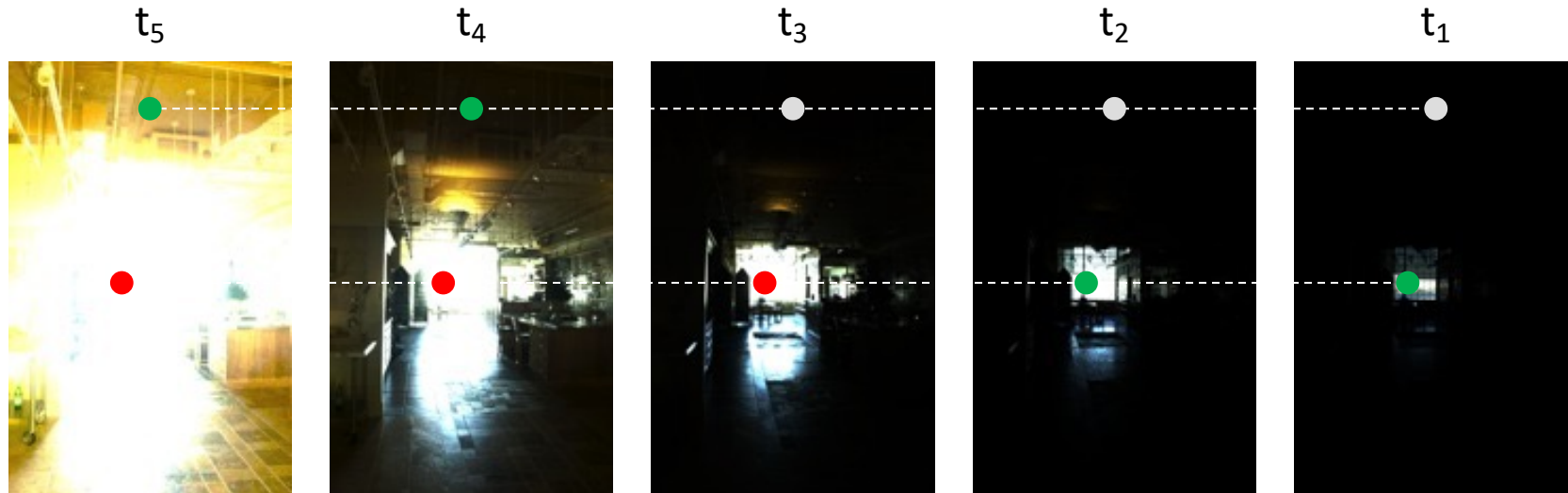


Merging images

For each pixel:

1. Find “valid” pixels in each image ← (noise) $0.05 < \text{pixel} < 0.95$ (clipping)
2. Weight valid pixel values appropriately
3. Form a new pixel value as the weighted average of valid pixel values

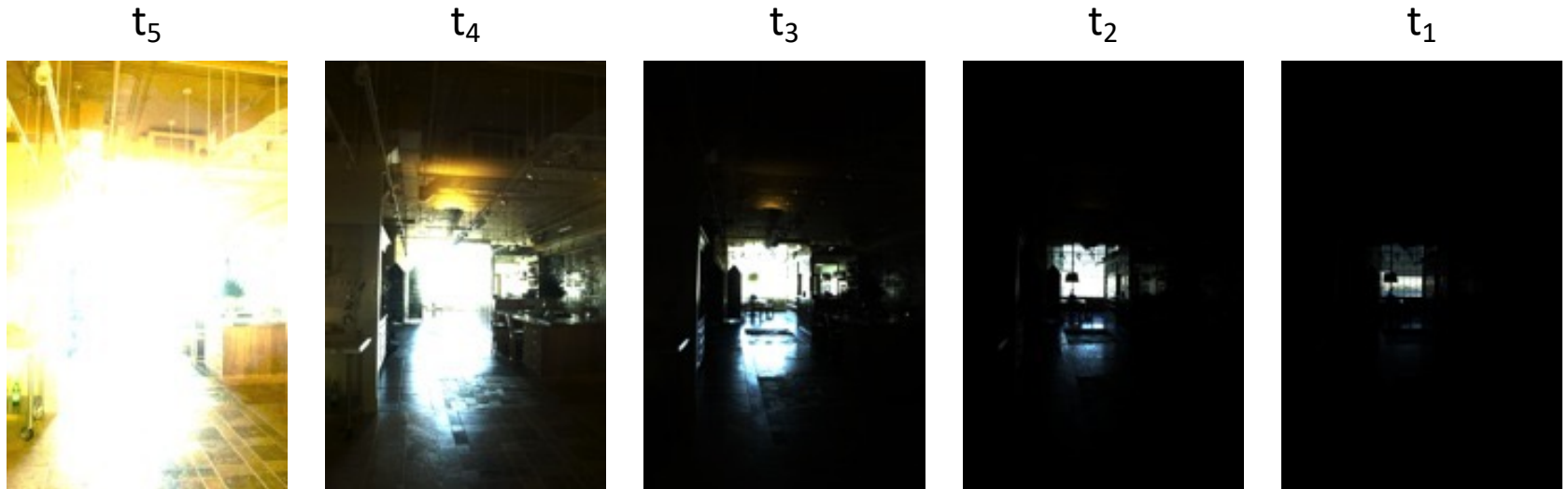
● noise
● valid
● clipped



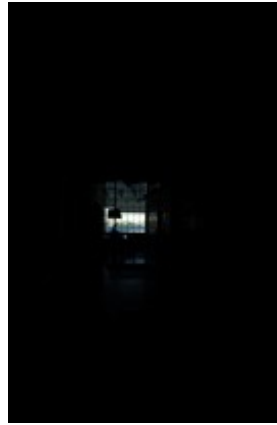
Merging images

For each pixel:

1. Find “valid” pixels in each image
2. Weight valid pixel values appropriately $\longleftarrow (\text{pixel value}) / t_i$
3. Form a new pixel value as the weighted average of valid pixel values



Merging result



Today

High Dynamic Range Imaging (HDR)

Deblurring

Colorization

Super Resolution

Reason of blurring

Defocus: the subject is not in the depth of view



Reason of blurring

Motion blur: moving subjects or unstable camera

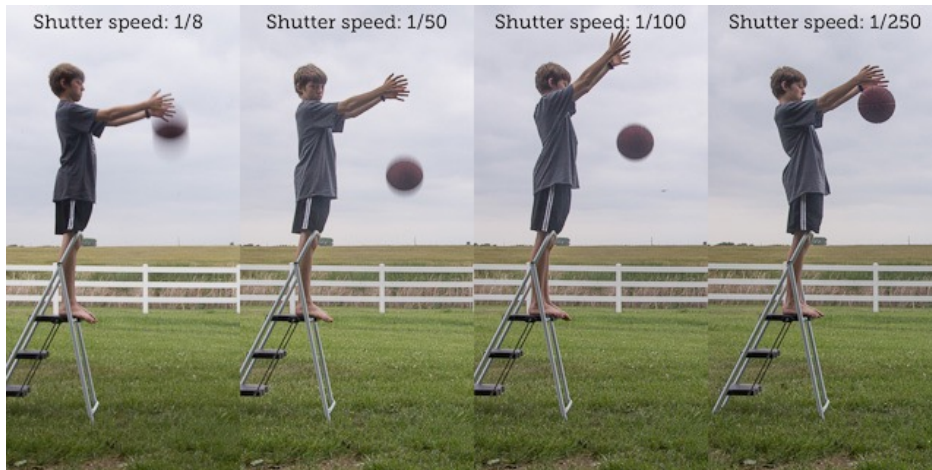


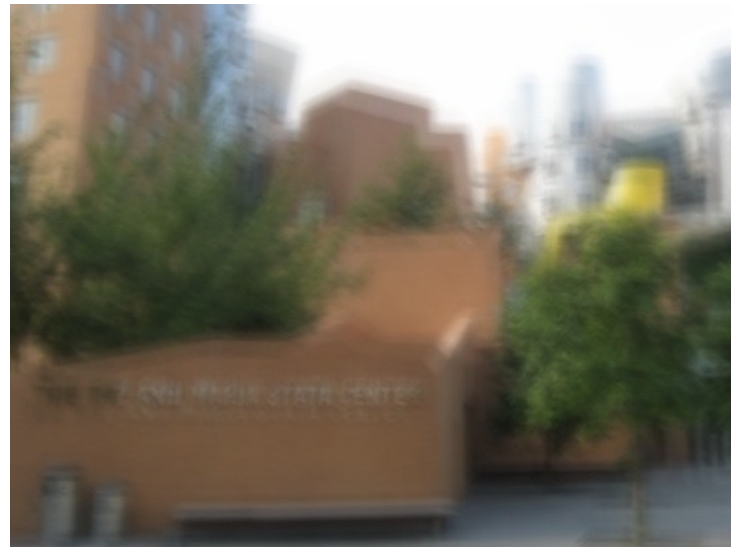
Image deblurring

Can we design an algorithm to solve this problem?



Modeling image blur

How to describe the image blur with a mathematical model?







Modeling image blur

- The blurring process can be described by convolution
- The blurred image is called convolution kernel

0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0

$F[x, y]$



a	b	c
d	e	f
g	h	i

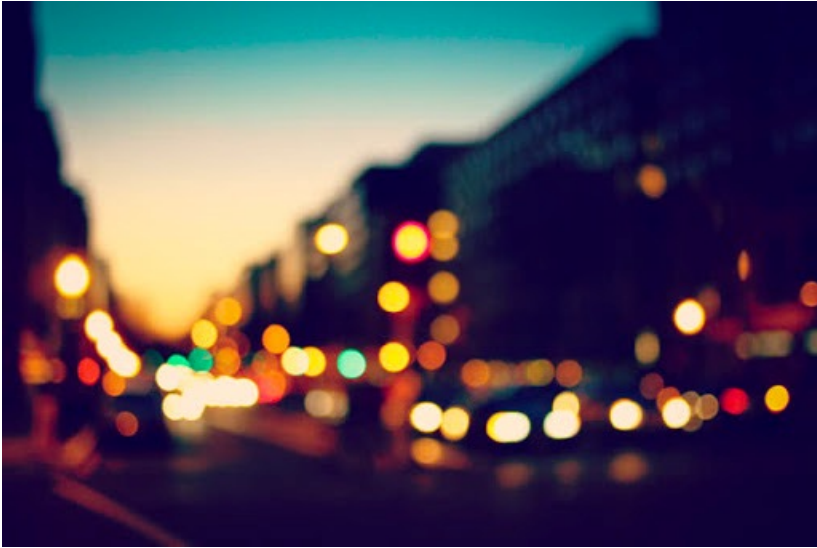
$H[u, v]$

=

a	b	c		a	b	c
d	e	f		d	e	f
g	h	i		g	h	i
a	b	c		a	b	c
d	e	f		d	e	f
g	h	i		g	h	i

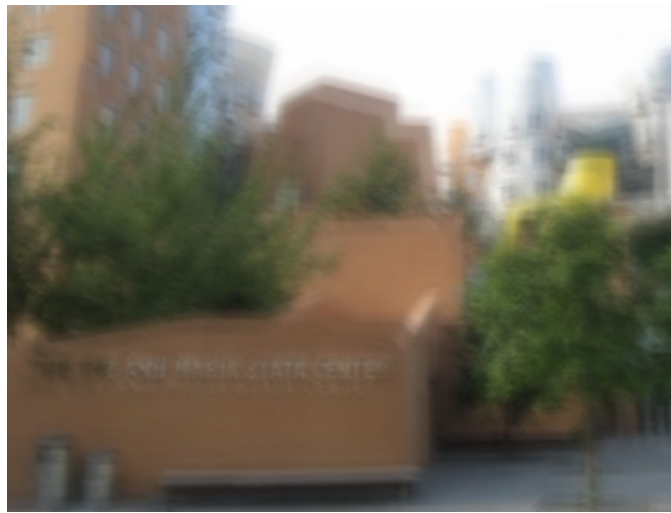
$G[x, y]$

Modeling image blur



The blur pattern of defocusing depends on the aperture shape
The blur pattern of shaking depends on the camera trajectory

Modeling image blur

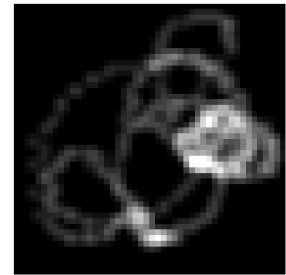


**blurred image
(input)**

=



**clear image
(to recover)**



**blur
kernel**

convolution

Deblurring = Deconvolution



Non-blind image deconvolution,
NBID



Blind image deconvolution,
BID

Non-blind image deconvolution (NBID)



$$G = F \otimes H$$

- G : The captured image (known)
- F : Image to be solved (unknown)
- H : Convolution kernel (known)

Frequency domain deconvolution

Convolution in the spatial domain = product in the frequency domain

$$FFT(G) = FFT(F \otimes H) = FFT(F) \times FFT(H)$$

Spatial deconvolution = division in frequency domain

$$F = IFFT(FFT(G) \div FFT(H))$$

Usually called inverse filter



Example

Guassain blur

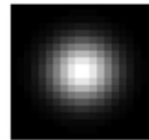
original



$f(x,y)$



$$\frac{1}{2\pi\sigma^2}e^{-r^2/2\sigma^2}$$



$h(x,y)$



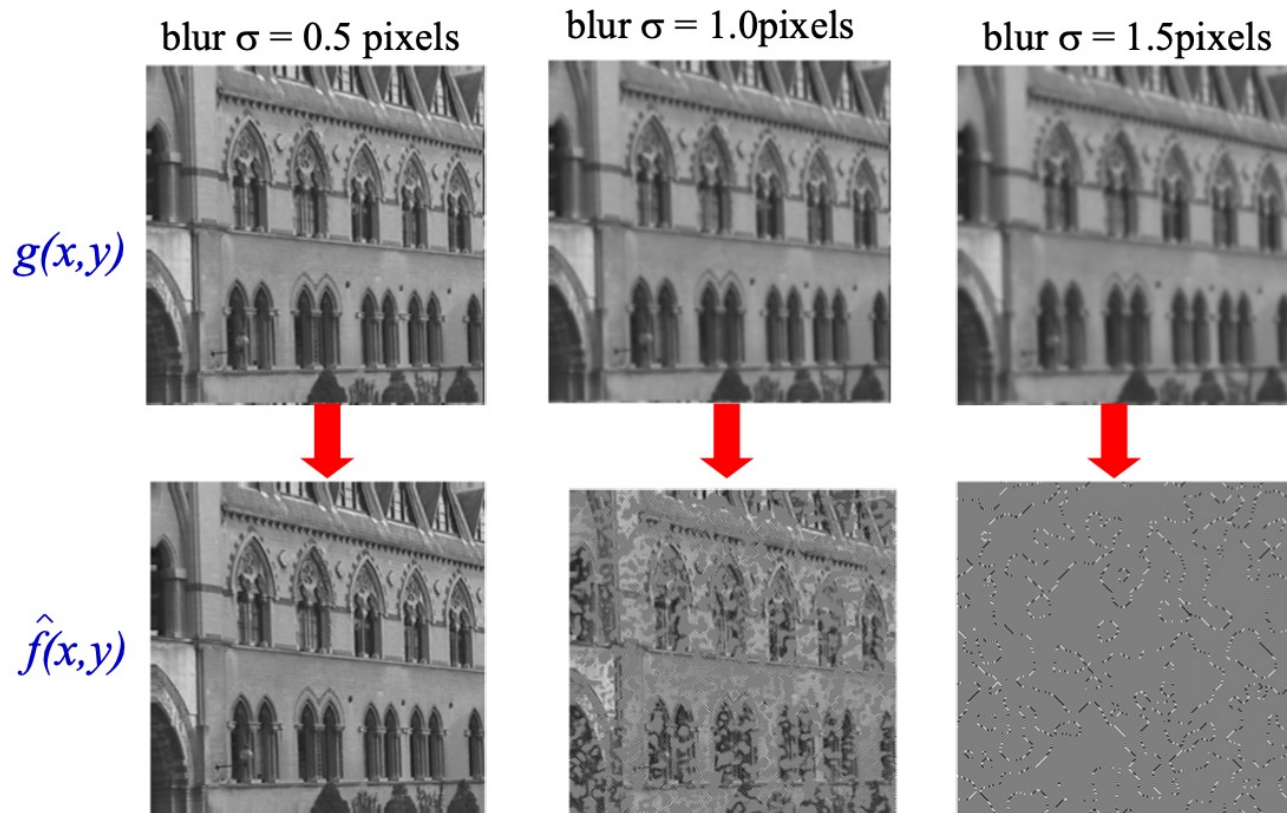
blurred



$g(x,y)$

Example

Inverse filtering results



Frequency domain deconvolution

The blurred kernel is generally a low-pass filter

Inverse filter will amplify high frequency components



Inverse filter will also amplify noise

$$G(u,v) = H(u,v) F(u,v) + N(u,v)$$

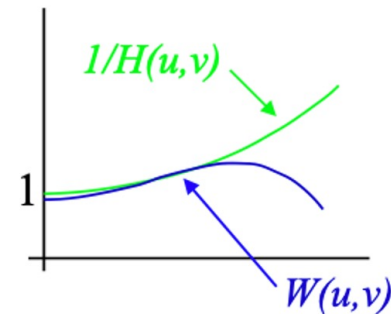
$$\hat{F}(u,v) = G(u,v) / H(u,v) = F(u,v) + N(u,v) / H(u,v)$$

Wiener Filter

Suppress high frequency when reverse filtering

$$\hat{F}(u, v) = W(u, v) G(u, v)$$

$$W(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K(u, v)}$$



- If $K = 0$ then $W(u, v) = 1 / H(u, v)$, i.e. an inverse filter
- If $K \gg |H(u, v)|$ for large u, v , then high frequencies are attenuated
- K is set to a constant scalar which is determined empirically

Example

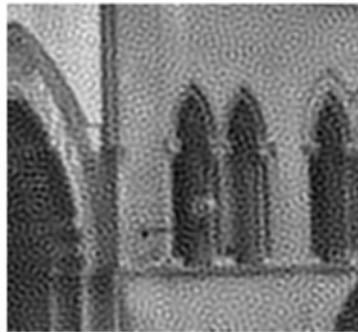
$g(x,y)$



$\hat{f}(x,y)$



$K = 1.0 \text{ e } -5$



$K = 1.0 \text{ e } -3$



$K = 1.0 \text{ e } -1$



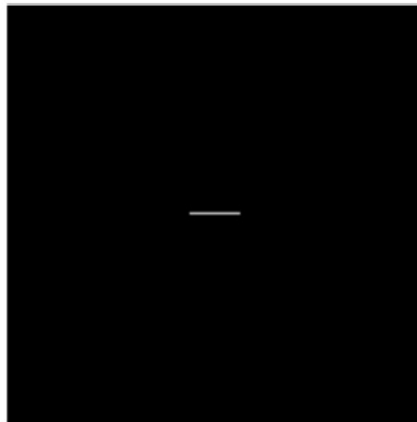
Application



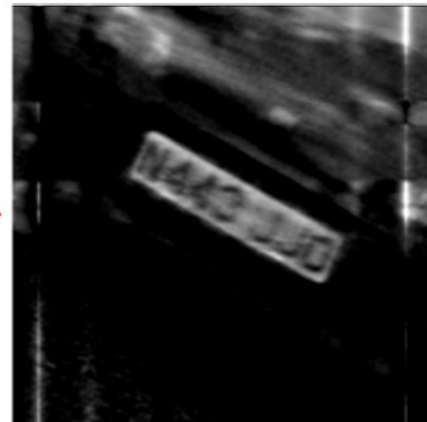
$f(x,y)$



$h(x,y)$



$\hat{f}(x,y)$



Application

Hubble Space Telescope



Image of star



Application

Hubble Space Telescope

Before and after corrective optics



http://en.wikipedia.org/wiki/Hubble_Space_Telescope

Any other method?

We can solve deconvolution by optimization

Variables to be optimized?

- Image to be recovered

Objective function?

- The similarity of the blurred image and the given blurred image (likelihood)
- The recovered image looks real (prior)

Deconvolution by optimization

Blurred image generation process:

$$G = F \otimes H + N$$

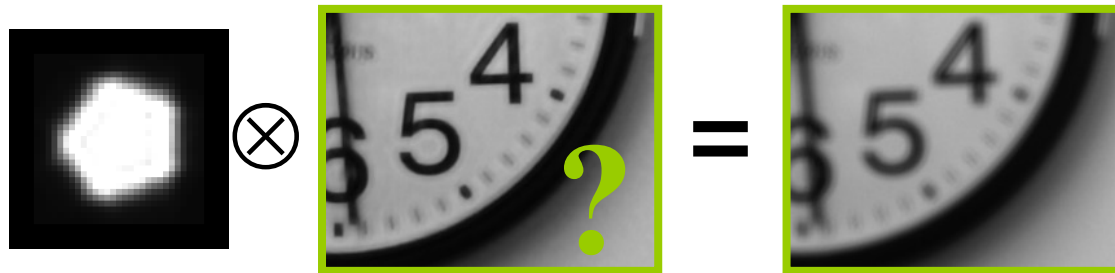
Assuming that the noise N is Gaussian noise, the likelihood can be measured by the mean square error:

$$MSE = \| G - F \otimes H \|_2^2 = \sum_{ij} (G_{ij} - [F \otimes H]_{ij})^2$$

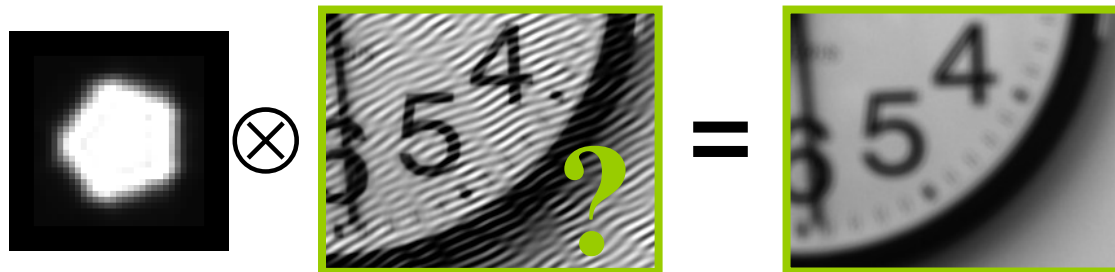
Deconvolution is ill-posed

Non-unique solution:

Solution 1:



Solution 2:

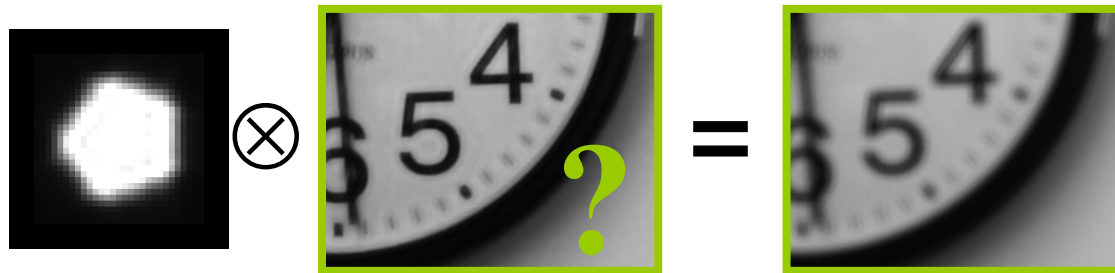


These two sets of solutions have the same MSEMSE

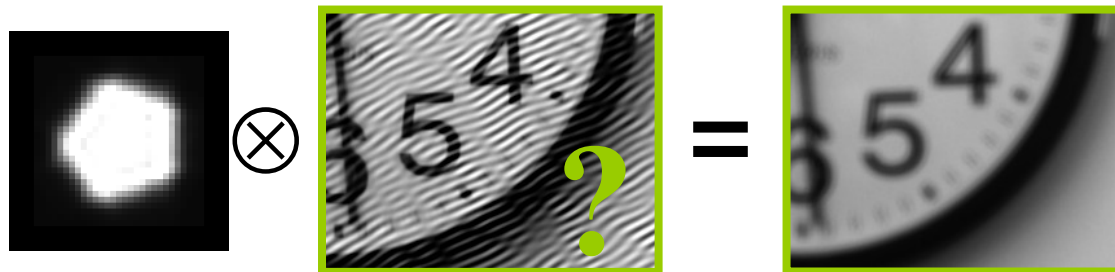
Deconvolution is ill-posed

Need prior information about the solution

Solution 1:



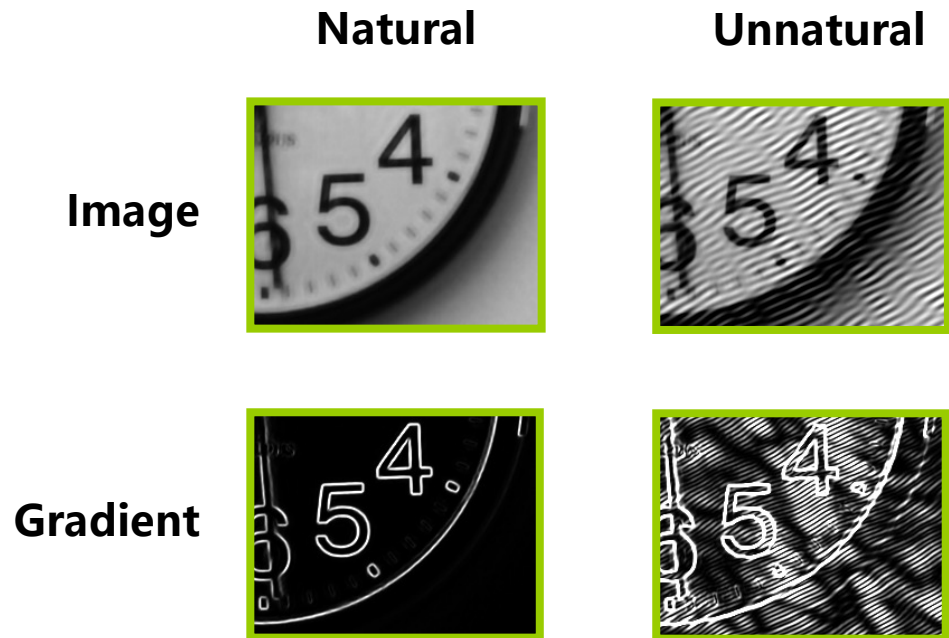
Solution 2:



Solution 1 looks more natural

Prior of natural image

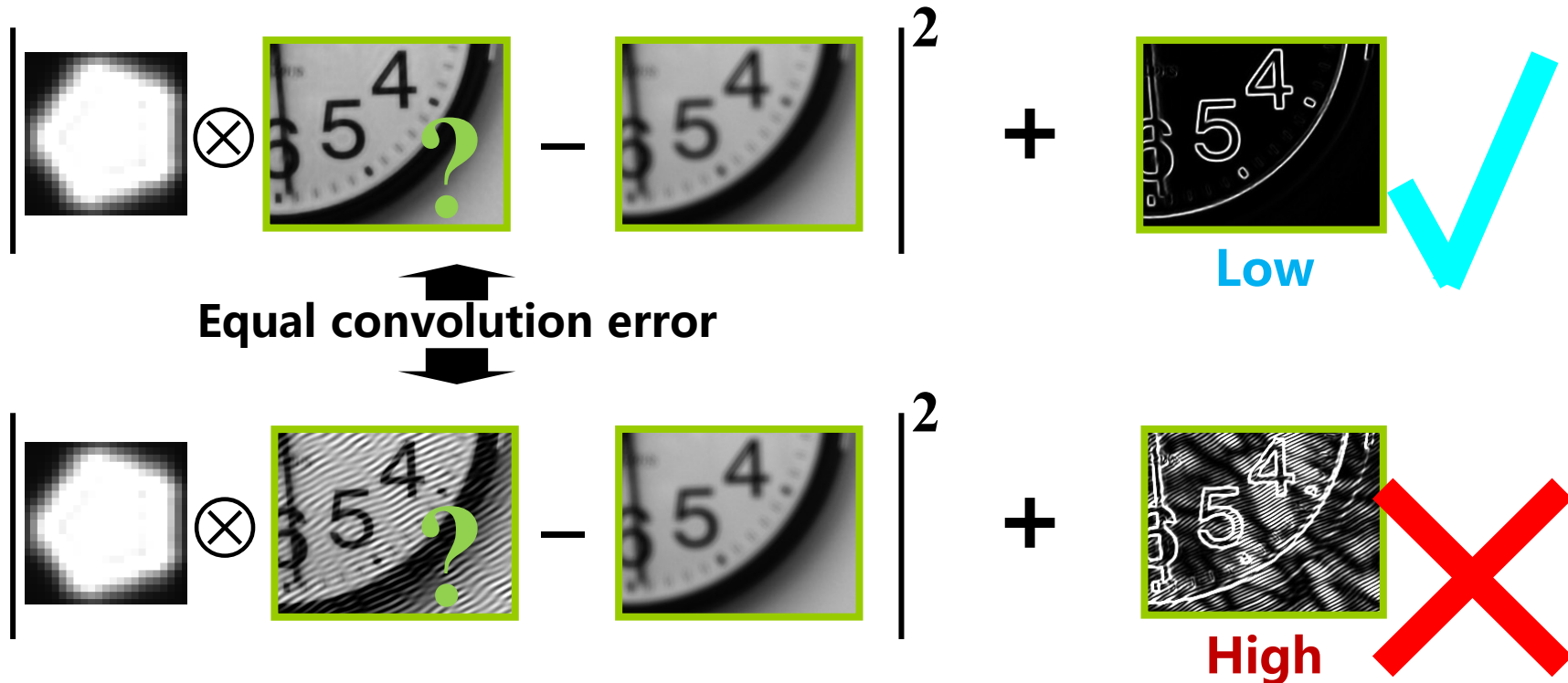
- Natural images are generally smooth in segments
- Gradient map is sparse
- Adding L1 regularization makes the image gradient sparse



Deconvolution with prior

Objective function = likelihood function + regular term

$$\min_F \| G - F \otimes H \|_2^2 + \| \nabla F \|_1$$



Blind Image Deconvolution (BID)

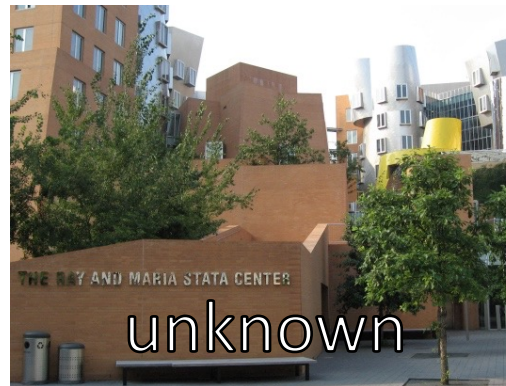
The convolution kernel is also unknown

Obviously more difficult

Need more prior knowledge!



=

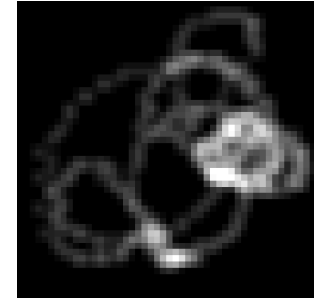


$\otimes$



Kernel prior

Blur kernel is non-negative and sparse



Optimized objective function:

$$\min_{F,H} \| G - F \otimes H \|_2^2 + \lambda_1 \| \nabla F \|_1 + \lambda_2 \| H \|_1$$

$$s.t. H \geq 0$$

Removing Camera Shake from a Single Photograph

Rob Fergus, Barun Singh, Aaron Hertzmann,
Sam T. Roweis and William T. Freeman

SIGGRAPH 2006

Massachusetts Institute of Technology
and
University of Toronto

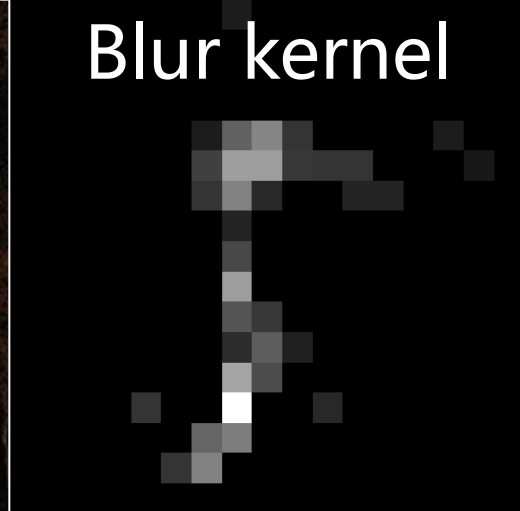
Original photograph



Output



Blur kernel



Today

High Dynamic Range Imaging (HDR)

Deblurring

Colorization

Super Resolution

Until 1970, color photos were still very rare, and people could only look back on history through black and white photos.



Nowadays, it is easy to use technology to change black and white to color!



Colorization

- Colorization refers to the process of adding color to a monochrome picture or video with the aid of a computer
- There are two main ways to color grayscale images:
 - Sample-based colorization: use sample image
 - Interactive colorization: paint brush interactively

Sample-based colorization



Source image

+



Target image

=



Result

Welsh, Tomihisa, Michael Ashikhmin, and Klaus Mueller. "Transferring color to greyscale images." *Proceedings of the 29th annual conference on Computer graphics and interactive techniques*. 2002.

Sample-based colorization



Source image

+



Target image

=



Result

Scan the target image, for each pixel:

- Find the best matching point in the sample (e.g., considering the brightness and the standard deviation with neighboring pixels)
- Assign the color of the matching point to the pixel

Interactive colorization

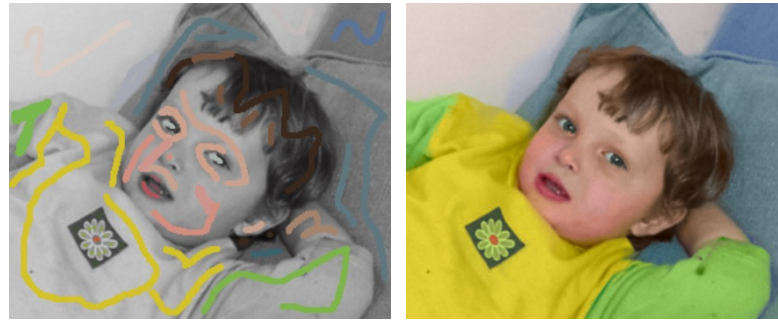


Input: gray image with brush



output

Interactive colorization



For two adjacent pixels, if the brightness is similar, then the color should also remain similar

Based on this assumption, the coloring problem is transformed into an optimization problem:

$$J(U) = \sum_r \left(U(r) - \sum_{s \in N(r)} w_{rs} U(s) \right)^2$$

Interactive colorization

Objective function:

$$J(U) = \sum_r \left(U(r) - \sum_{s \in N(r)} w_{rs} U(s) \right)^2$$

$U(r), U(s)$: RGB values of pixel r, s

$N(r)$: neighborhood pixels of pixel r

w_{rs} : Weight that measures similarity between r and s

Constraint:

User-specified colors of brushed pixels keep unchanged

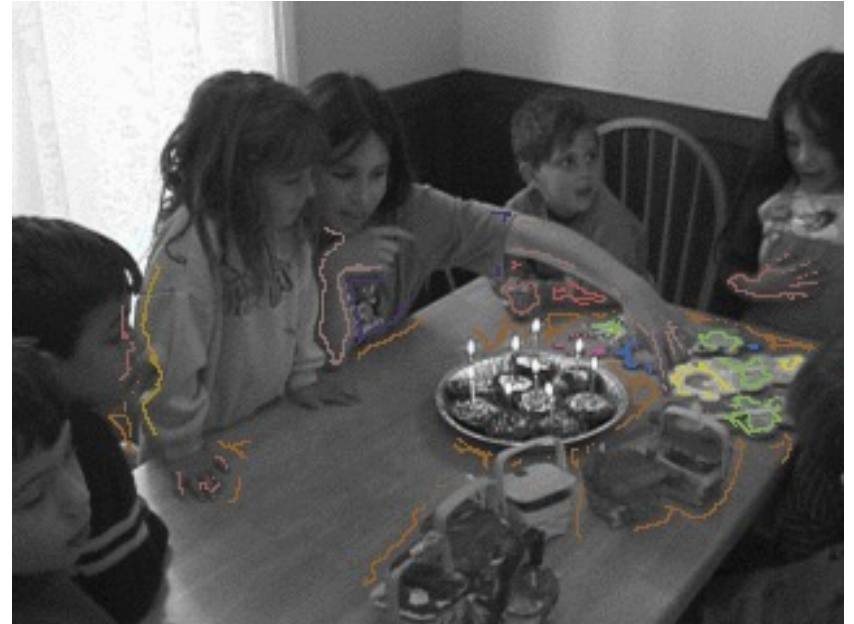
Results



Video colorization



Original video (62fps)



Brush (10fps)

Video colorization



Input



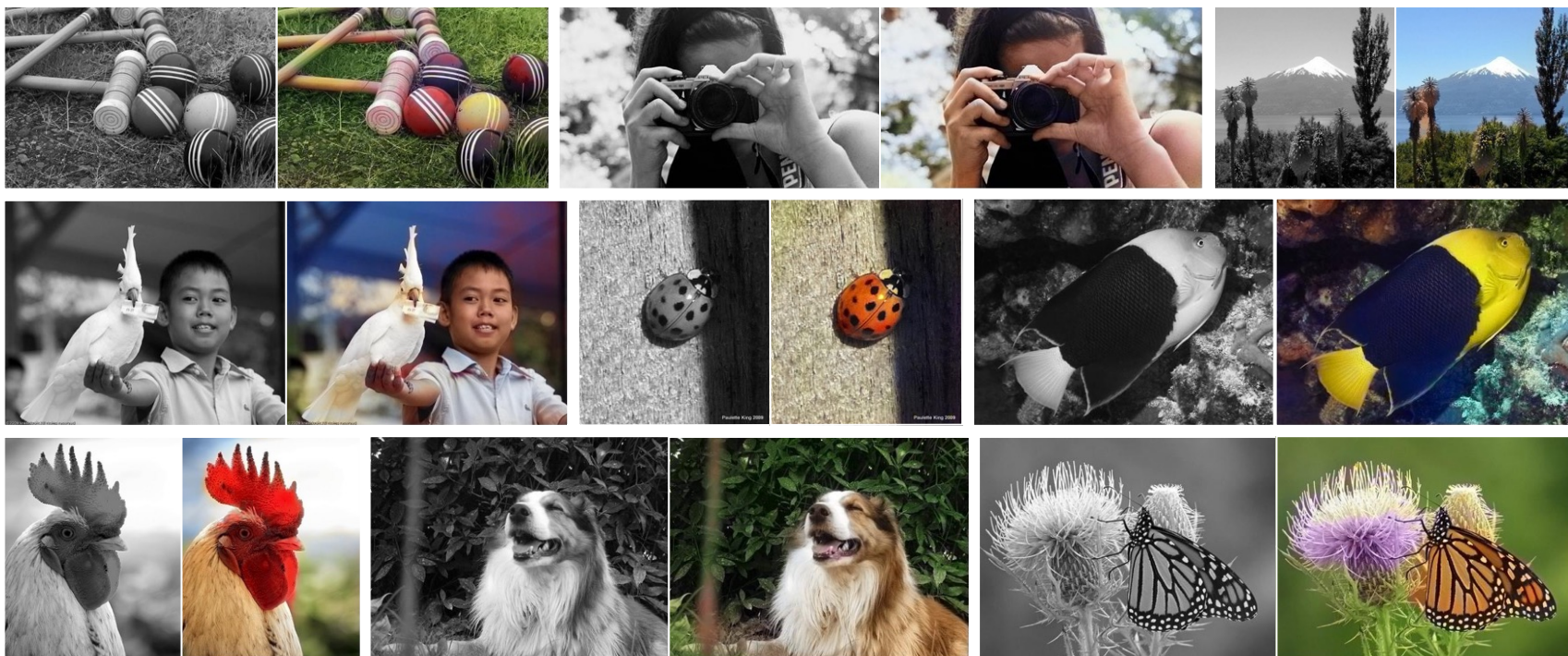
Output

The modern approach?

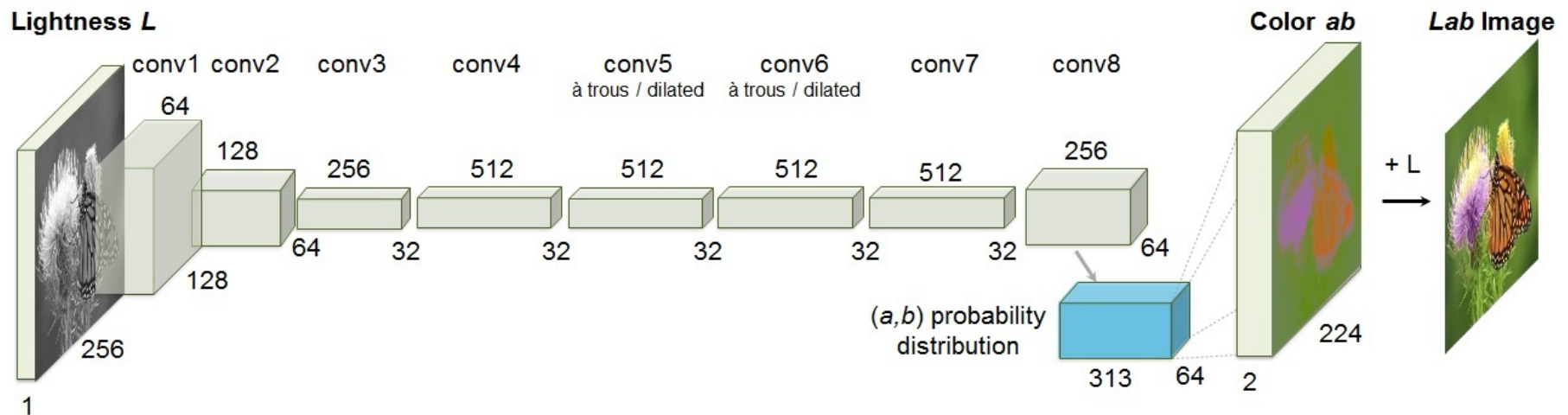
Colorful Image Colorization

Richard Zhang, Phillip Isola, Alexei A. Efros
`{rich.zhang,isola,efros}@eecs.berkeley.edu`

University of California, Berkeley



Convolutional Neural Networks



<http://richzhang.github.io/colorization/>

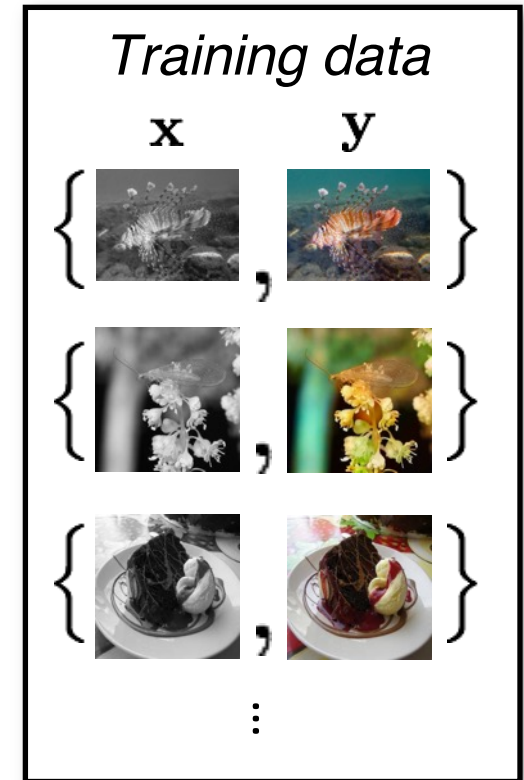
Loss function for image synthesis

Reconstruction loss

$$L(\Theta) = ||F(X; \Theta) - Y||^2$$

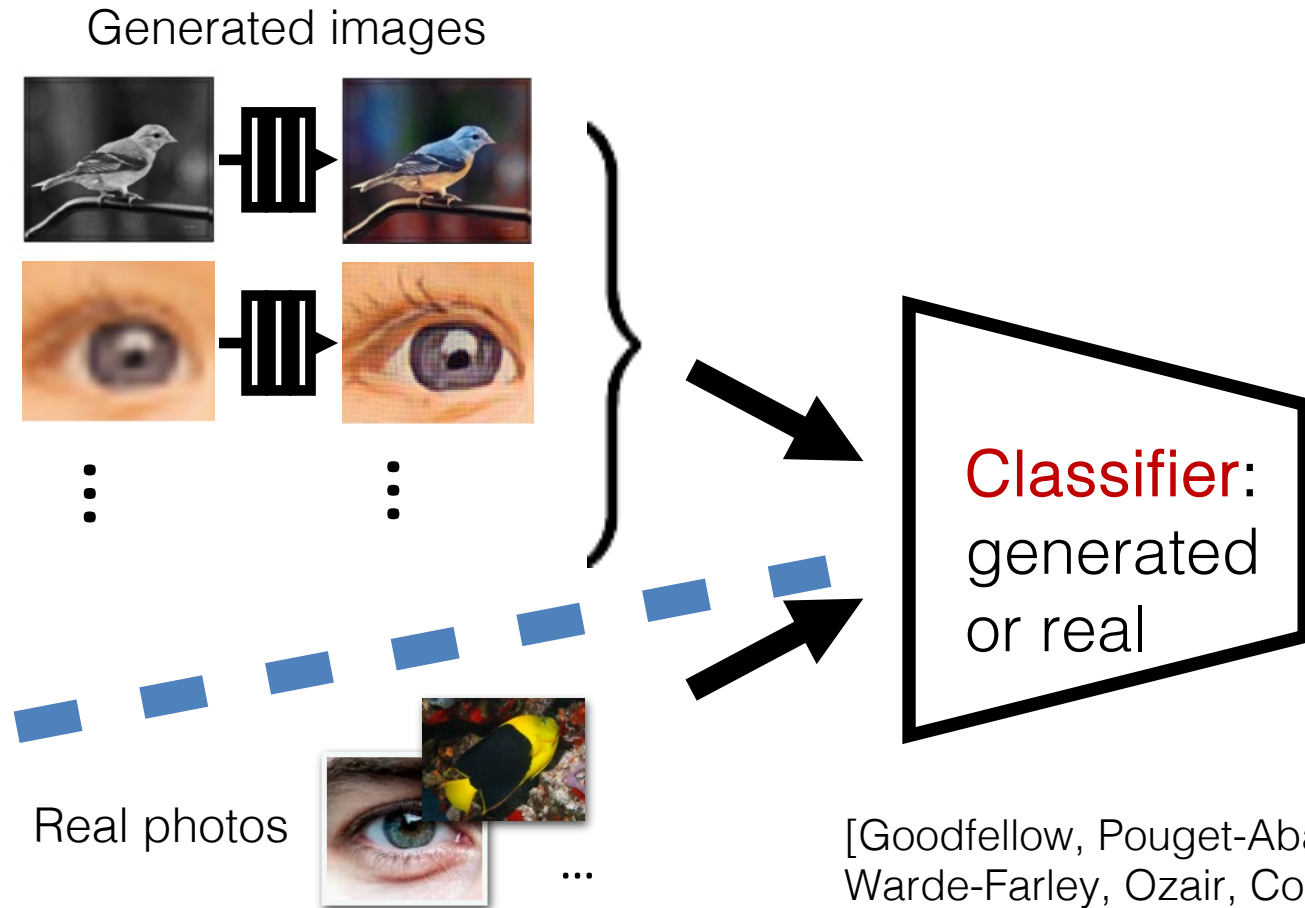
Problem with reconstruction loss

- Cannot handle the case with multiple solutions
- Cannot measure if an image is realistic

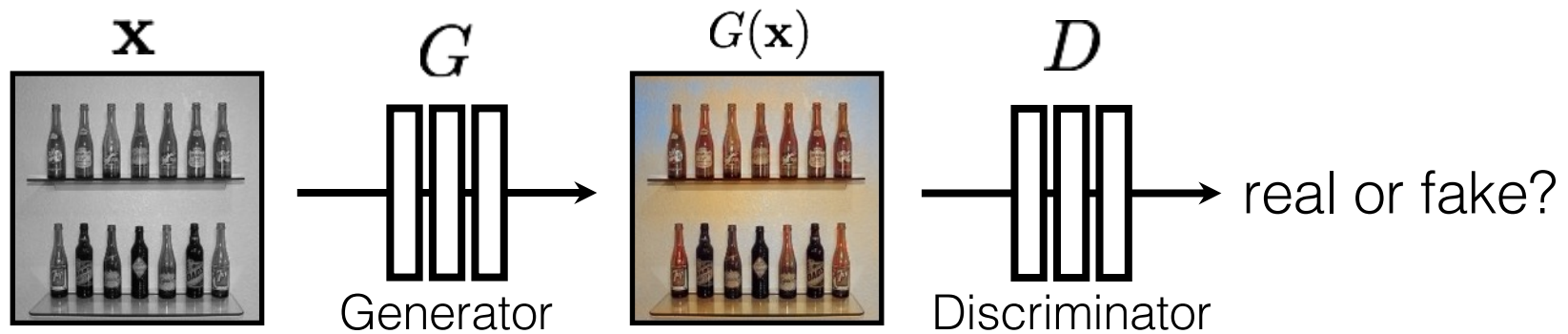


Can we learn a loss function that measures if an image is real?

Generative Adversarial Network (GAN)



Generative Adversarial Network (GAN)

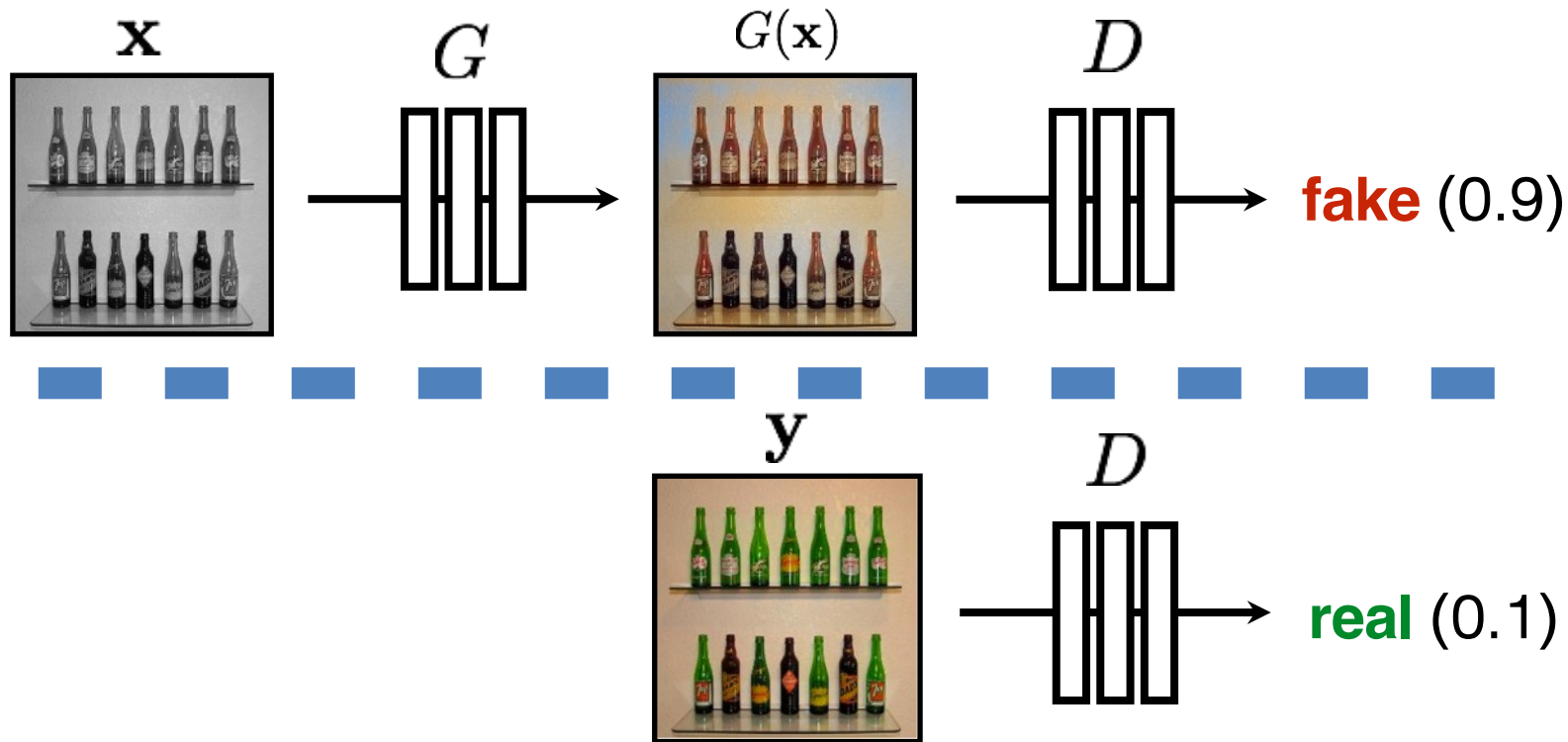


G tries to synthesize fake images that fool **D**

D tries to identify the fakes

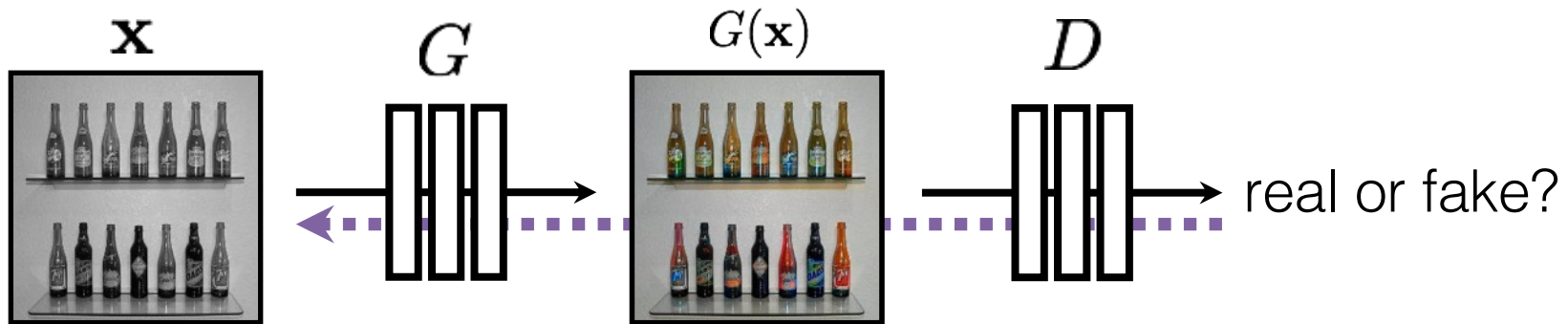
[Goodfellow et al., 2014]

Generative Adversarial Network (GAN)



$$\arg \max_D \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(G(\mathbf{x})) + \log(1 - D(\mathbf{y}))]$$

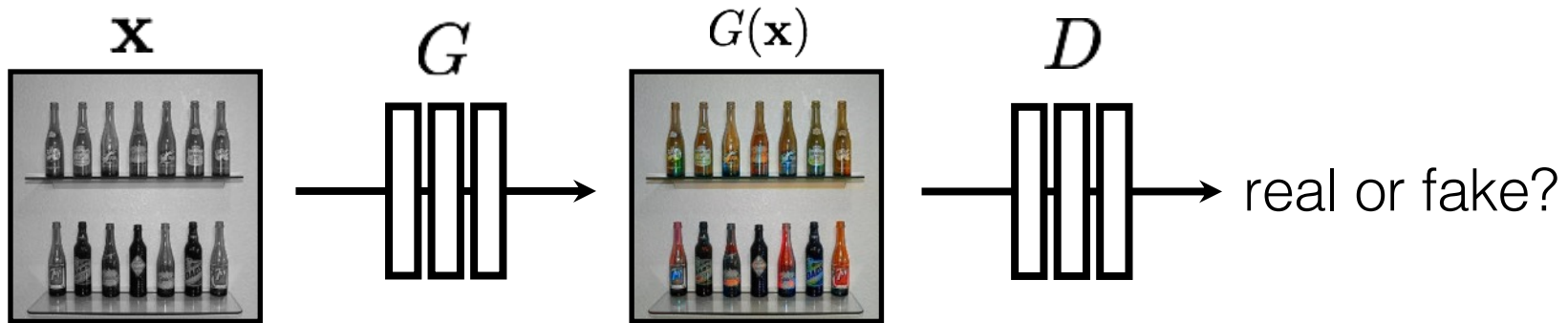
Generative Adversarial Network (GAN)



G tries to synthesize fake images that *fool* **D**:

$$\arg \min_G \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(G(\mathbf{x})) + \log(1 - D(\mathbf{y}))]$$

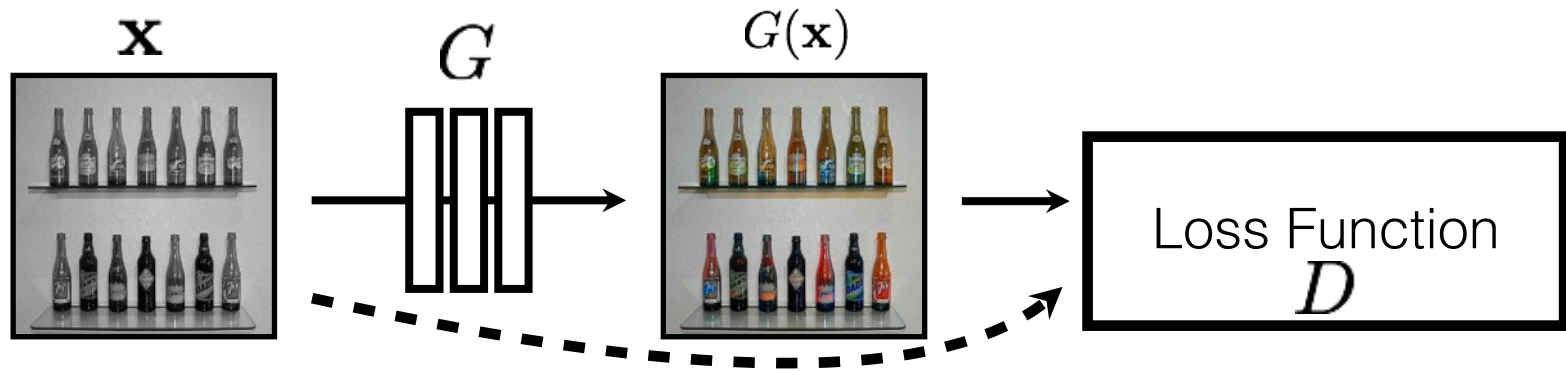
Generative Adversarial Network (GAN)



G tries to synthesize fake images that *fool* the *best* **D**:

$$\arg \min_G \max_D \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(G(\mathbf{x})) + \log(1 - D(\mathbf{y}))]$$

Generative Adversarial Network (GAN)



$$\arg \min_G \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(G(\mathbf{x}))]$$

D can be viewed as a loss function to train G

- Called **adversarial loss**
- Learned instead of being hand-designed
- Can be applied to any image synthesis tasks

[Goodfellow et al., 2014]

[Isola et al., 2017]

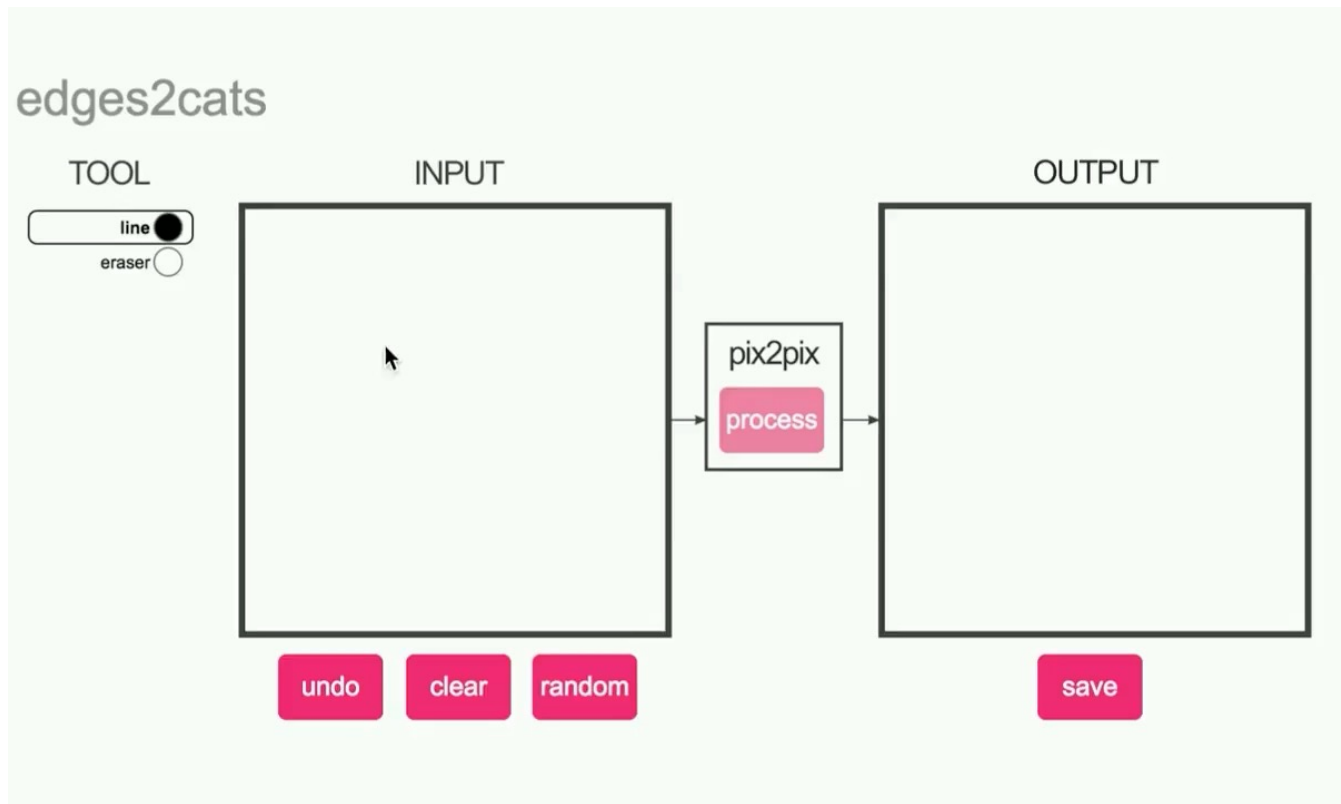
More image synthesis tasks

Image to Image Translation



More image synthesis tasks

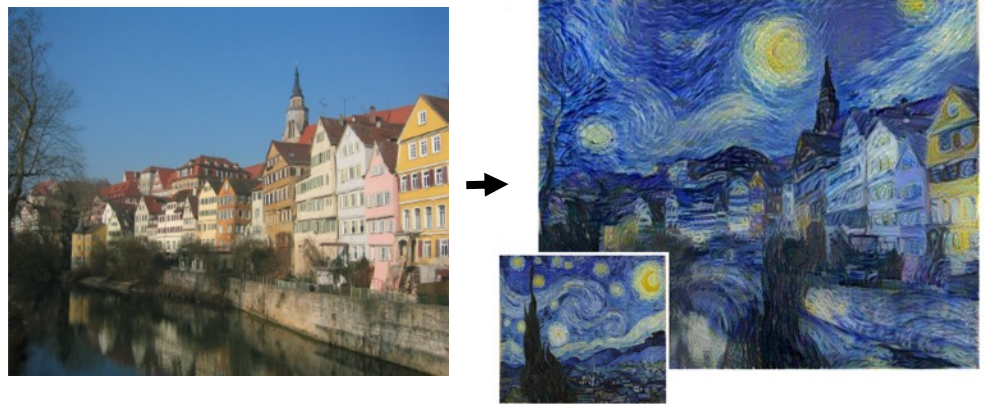
Image to Image Translation



<https://affinelayer.com/pixsrv/>

More image synthesis tasks

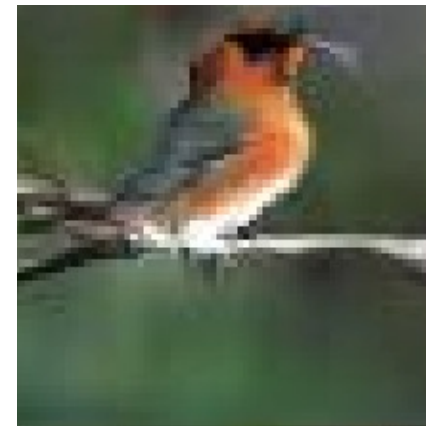
Style transfer



[Gatys et al. 2016, ...]

Text-to-Photo

“this small bird has a pink breast
and crown...”



[Reed et al. 2014, ...]

More image synthesis tasks

Image dehazing



Cycle-Dehaze: Enhanced CycleGAN for Single Image Dehazing. CVPRW 2018
Deniz Engin*, Anıl Genc*, Hazım Kemal Ekenel

Today

High Dynamic Range Imaging (HDR)

Deblurring

Colorization

Super Resolution

Super-Resolution



Original

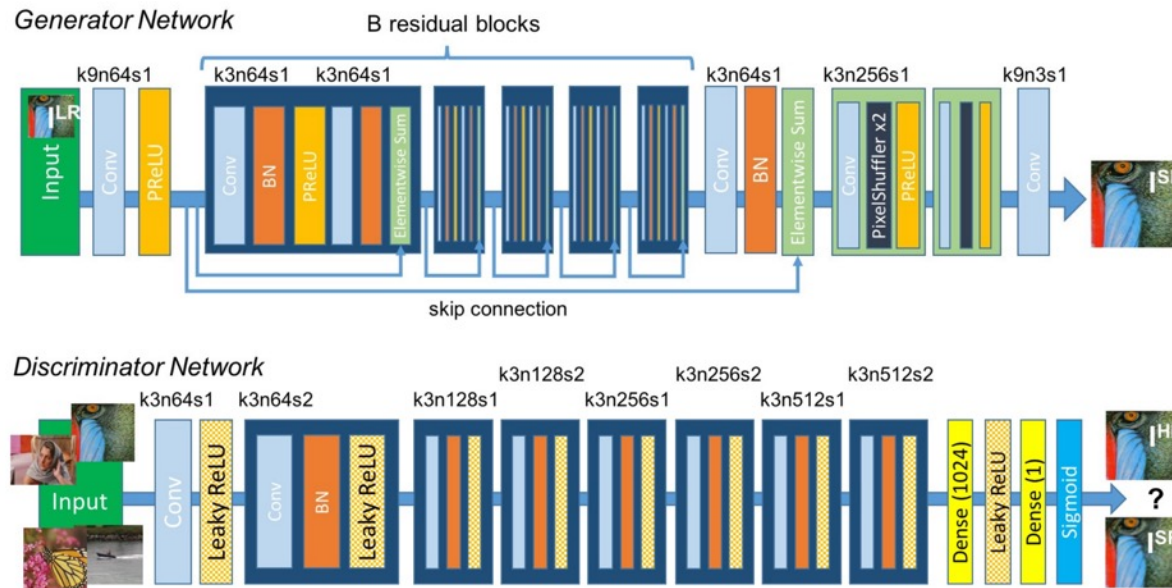


Bi-Cubic



Super-Resolution

Super Resolution using GAN



$$l^{SR} = \underbrace{l_X^{SR}}_{\text{reconstruction loss (likelihood)}} + 10^{-3} \underbrace{l_{Gen}^{SR}}_{\text{adversarial loss (regularizer)}}$$

$$\sum_{x=1}^{rW} \sum_{y=1}^{rH} (I_{x,y}^{HR} - G_{\theta_G}(I_{x,y}^{LR}))^2 \quad \sum_{n=1}^N -\log D_{\theta_D}(G_{\theta_G}(I^{LR}))$$

Super Resolution using GAN



Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]

Question?